

Face Mask Detection with Live Alert System

Problem Statement

Manual enforcement of mask-wearing guidelines in public and private spaces is inefficient and unreliable. This project addresses the lack of automated, real-time monitoring by developing a computer vision-based system that detects face masks in live video streams and triggers an immediate alert when a person is unmasked.

Abstract

This project implements a real-time Face Mask Detection system using Deep Learning and Computer Vision techniques. The system detects whether a person is wearing a mask using a trained Convolutional Neural Network (CNN) model. It integrates OpenCV for live webcam streaming, Haar Cascade for face detection, and Flask for web deployment. When a person is detected without a mask, the system triggers alerts including sound notifications and screenshot capture. The project demonstrates practical application of AI for public safety monitoring.

Introduction

With the increasing importance of health safety measures, automated mask detection systems became essential in public places such as offices, airports, and educational institutions. This project aims to build an AI-based monitoring system capable of detecting mask compliance in real-time using a webcam. The system processes video frames, detects faces, classifies mask usage, and generates alerts when violations occur.

Tools Used

- Python – Core programming language
- TensorFlow / Keras – Building and training CNN model
- OpenCV – Image processing and face detection
- Haar Cascade Classifier – Face detection algorithm
- Flask – Web application deployment
- NumPy – Numerical computations

Steps Involved in Building the Project

- Collected and organized dataset into 'with_mask' and 'without_mask' categories.
- Preprocessed images (resizing to 100x100 pixels and normalization).
- Built and trained a CNN model using Keras.
- Saved the trained model (.h5 file).
- Integrated model with OpenCV for real-time webcam detection.
- Implemented alert mechanism (sound alert and screenshot saving).
- Deployed the system using Flask with an aesthetic web interface.

Conclusion

The Face Mask Detection with Live Alert System successfully demonstrates the integration of Deep Learning and Computer Vision in real-time monitoring applications. The system efficiently detects mask compliance and generates alerts when necessary. This project highlights how AI can contribute to public health and safety while showcasing practical deployment skills using Flask.